

## WEEK-11—Amazon LEX

### Step 1: Open Amazon Lex

1. Login to AWS Console
2. Search **Amazon Lex**
3. Click **Create bot**

### Step 2: Create Bot

1. Choose:
  - **Create a blank bot**
2. Enter:
  - Bot name: HotelBookingBot
3. IAM role → Create new role
4. Remaining default → next
5. Language: English
6. Voice (optional)
7. Click **Done**

### Step 3: Create Intent

1. Go to **Intents**
2. Name: BookHotel
3. Click **Create**

### Step 4: Add Sample Utterances

Add user sentences (how user talks):

I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**

### Step 5: Create Slots (User Inputs)

Slots collect user information.

#### Add Slot

1. Click **Add slot**
2. Fill:
  - Slot name: age
  - Slot type: AMAZON.Number
  - Prompt: *Which city do you want?*

Mark as **Required**

#### Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

**Example Condition:**

If {age} < 18

**Steps:**

1. Go to **Intent** → **Slots** → under **Advance options**
2. Scroll to **Slot capture: success response** → **Conditional branching**
3. Click **Add condition**

**Example:****Condition:**

If {age} < 18

**Response:**

Your not eligible for hoot booking?

**Step 6: Create Slots (User Inputs)**

Slots collect user information.

**Add Slot**

3. Click **Add slot**
4. Fill:
  - Slot name: location
  - Slot type: AMAZON.City
  - Prompt: *Which city do you want?*
5. Mark as **Required**

Repeat for other slots.

1. Click **Add slot**
2. Fill details:
  - **Slot name:** checkin
  - **Slot type:** AMAZON.Date
  - **Prompt:**  
*"What is your check-in date?"*
3. Mark as **Required**  
(Optional but recommended)
  - Add **Retry prompt:**  
*"Please provide a valid date (e.g., 2026-03-25)"*
4. Click **Save**

**Slot 7: Number of Nights (nights)****Steps:**

1. Click **Add slot** again
2. Fill details:
  - **Slot name:** nights
  - **Slot type:** AMAZON.Number

- **Prompt:**  
*"How many nights will you stay?"*
- 3. Mark as **Required**
- 4. Click **Save**

## Step 7: Create Custom Slot Type

### Save intent

#### Click on Back to intents

If you want custom values:

1. Go to **Slot Types**
2. Click **Add slot type** -> **add blank slot type**
3. Name: RoomType

Add values:

Single

Double

Suite

Save slot type

Now use this in slot: go back to intent

- Slot name: RoomType
- Slot type: RoomType

## Step 9: Add Response Cards (Card Groups)

Used for buttons (quick replies)

### Steps:

1. Go to **slot prompts** -> **more prompt options**
2. Click **Add** -> **add card groups**

### Example:

**Card Title:** Select Room Type

**click on add Buttons:**

- Single → "Single"
- Double → "Double"
- Suite → "Suite"

## Step 10: Configure Responses

### Initial Response:

Welcome to Hotel Booking! What is your name?

### Confirmation:

Do you want to confirm booking in {location} for {nights} nights?

### Step 11: Build & Test

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:

Book a hotel

#### Sample output

User: Book a hotel

Bot: What is your age?

User: 25

Bot: Which city do you want?

User: Hyderabad

Bot: Check-in date?

User: Tomorrow

Bot: Number of nights?

User: 2

Bot: Select room type

User: Double

Bot: Do you want to confirm booking in Hyderabad for 2 nights?

User: Yes

Bot: Booking confirmed.

## SCENARIO BASED QUESTIONS

### Scenario 1: Bot Not Responding

You created a chatbot, but when you test it, the bot does not respond to user input.

- What could be the possible reason for no response?
- Did you build and publish the bot correctly?
- How would you troubleshoot this issue?

### Scenario 2: Incorrect Intent Matching

A user types a valid query, but the chatbot responds with the wrong intent.

- What could cause incorrect intent recognition?
- How do utterances affect intent matching?

- How can you improve accuracy?

### Scenario 3: Missing Slot Values

The chatbot asks for user input, but fails to capture required slot values.

- What is a slot in Amazon Lex?
- Why might slot values not be captured properly?
- How would you fix slot configuration issues?

### Scenario 4: Fulfillment Not Working

The bot collects all user inputs but does not perform the expected action.

- What is fulfillment in Amazon Lex?
- Could Lambda integration be misconfigured?
- How would you verify the backend processing?

### Scenario 5: Bot Not Accessible

You created the bot successfully, but users are unable to access it.

- Did you deploy or publish the bot?
- Could permissions or channel integration be missing?
- How would you make the bot accessible to users?

### Scenario 6: Wrong Language Understanding

The chatbot fails to understand user queries written in slightly different formats.

- How does Amazon Lex process natural language?
- Why is training with multiple utterances important?
- How can you improve NLP performance?

### Scenario 7: Versioning Issue

You updated the bot, but users still experience the old behavior.

- Did you create a new version of the bot?
- Why is alias configuration important?
- How do you ensure users access the latest version?

### Scenario 8: Lambda Permission Error

The chatbot is connected to a Lambda function, but it fails during execution.

- What type of permission error might occur?
- How are IAM roles used in this integration?
- How would you fix the permission issue?

### Scenario 9: Bot Gives Default Response

The chatbot always gives a fallback/default response instead of proper answers.

- What is a fallback intent?
- Why does the bot fail to match user input?
- How can you improve intent recognition?

### Scenario 10: Multi-turn Conversation Issue

The chatbot fails to continue conversation properly after asking follow-up questions.

- What is dialog management in Amazon Lex?
- How are slots used in multi-turn conversations?
- How can conversation flow be improved?

### Scenario 11: Integration with Web Application

A company wants to integrate the chatbot into their website.

- How can Amazon Lex be integrated with web applications?
- What services or APIs are required?
- What challenges might occur during integration?

#### Scenario 12: Performance Issue

The chatbot responds slowly when many users interact simultaneously.

- What could cause performance delays?
- How can scalability be handled in Amazon Lex?
- What AWS services can support high traffic handling?